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WORK EXPERIENCE

Research engineer

Special Telecommunications Service [27/07/2018 – Current]

City: Bucharest
Country: Romania

Design and development of cryptographic software.

University teaching assistant

Military Technical Academy "Ferdinand I" [02/2017 – 07/2021]

City: Bucharest
Country: Romania

Teaching and preparing labs on computer vision and machine learning.

EDUCATION AND TRAINING

Bachelor degree

Military Technical Academy "Ferdinand I" [2014 – 2018]

Field(s) of study: Computer Science
Final grade : 10
Thesis: Attacks on some cryptosystems using quantum computing

For my thesis, I've implemented quantum algorithms for key establishment (BB84, E91) and factorization (Shor's algorithm) using IBM-Q systems. I've also proposed a new quantum encryption scheme.

MSc degree

Military Technical Academy "Ferdinand I" [2019 – 2021]

Field(s) of study: Computer security
Final grade : 10
Thesis: Cloud based image processing using homomorphic encryption

For my thesis, I've studied fully homomorphic encryption (FHE) and I've implemented secure image processing algorithms using EVA (Compiler for Microsoft SEAL).

IBM AI Engineering

Coursera [2020 – 2021]

https://coursera.org/share/fac314e768a11bd80626cdd234802270

PhD

University of Bucharest [10/2021 – Current]

Field(s) of study: Computer Science

I'm studying neural cryptography in the framework of P-Systems. My main research interest is in the usage of neural networks in cryptography, especially tree parity machines synchronization for key agreement protocols.

LANGUAGE SKILLS

Mother tongue(s): **Romanian**

Other language(s):

English

LISTENING B2 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

PUBLICATIONS

A New Quantum Encryption Scheme

[2018]
https://journals.aijr.in/index.php/ajgr/article/view/699

The model of quantum computation has advanced very quickly in the last years. This model brings with it an efficient algorithm for factoring, namely the Shor algorithm. This means that the public key infrastructure will soon be obsolete. In this paper we propose a new quantum cryptographic scheme which aims to replace the RSA algorithm from current public key infrastructures. We analyze the security of our scheme and also, we describe the implementation of the scheme using IBM Q SDK, qiskit. We run a number of experiments in order to build a proof of concept application that uses the proposed scheme.

Hybrid scheme for secure communications using quantum and classical mechanisms

[2017]
https://www.researchgate.net/profile/Mihail_Iulian_Plesa/publication/321676348_Hybrid_scheme_for_secure_communications_using_quantum_and_classical_mechanisms/links/5c6568ca92851c48a9d3b622/Hybrid-scheme-for-secure-communications-using-quantum-and-classical-mechanisms.pdf

The paper proposes a new scheme for secure data transmission based on a hybrid technology quantum and classical. Our scheme addresses two important problems: the confidentiality and integrity of data. The scheme is based on quantum teleportation circuit but the data transmission is accomplished through classical channels. Several experiments were conducted using the new IBM Q platform.

A quantum safe analysis of Helios voting system

[2020]
https://acad.ro/sectii2002/proceedings/doc2020-1/07-Plesa.pdf

In this paper, we evaluate how quantum computers affect an electronic voting system in general but we also provide a detailed analysis of the Helios voting system. We analyze the two major cryptographic components of Helios that are vulnerable to quantum computers: the encryption scheme and the decryption proof. We investigate the use of Niederreiter cryptosystem as a replacement for the ElGamal encryption scheme and proposed a new decryption proof for this scheme.

Using quantum communications for maritime signal flags

[2019]
https://search.proquest.com/openview/86758f2a88a42567a5cfed7573cecd83/1?pq-origsite=gscholar&cbl=2036237

Quantum communications are becoming very quickly a reality. There are huge advancement made in the field of quantum internet. Recently, IBM has announced the first commercial quantum computer with 20 qubits. Given all the advancements in the field, in this paper we investigate how quantum technologies can be applied in maritime communications. In this paper we address the problem of international maritime flag signals. More exactly, we proposed some quantum communication schemes for international maritime signal flags. We are also study the efficiency and security boost that quantum communications give in this type of maritime communication.

Privacy-Preserving Clustering: A New Approach Based on Invariant Order Encryption

[2021]

Cloud computing is increasingly used. One main use of cloud computing is the running of a machine learning algorithm. Due to the large amount of data required for these algorithms, they can no longer be run on personal computers. Uploading personal data to the cloud automatically raises the issues of confidentiality of this data. In this paper, we show through a series of experiments that an order-preserving encryption algorithm can be applied to guarantee the confidentiality of the input of two well-known clustering algorithms: K-Means and DBSCAN. We show that K-Means can be modified to be applied over the encrypted data. We also proposed a slight improvement to an order-preserving encryption scheme to ensure that it is randomized, therefore increasing its security level. Finally, after studying the performance of clustering algorithms over encrypted data we show a practical application of this idea, namely the color reduction over an encrypted image.

DRIVING LICENCE

Driving Licence: B

CONFERENCES AND SEMINARS

DefCamp

[Bucharest, 2019]

How to implement a quantum key distribution protocol on IBM's quantum computer (a practical example of quantum programming)

https://www.youtube.com/watch?v=7o3vTOYTbNs

SOCIAL

Profiles

Google Scholar: https://scholar.google.ro/citations?user=dvlBs70AAAAJ&hl=ro

CClaim: https://www.youracclaim.com/users/mihail-iulian-plesa

PROJECTS

EMA

Designing and developing an end-to-end encrypted messaging app for iOS systems.

Responsibilities:

- Development of cryptographic protocols for e2ee one-to-one messaging and group messaging
- Development of cryptographic protocols for e2ee call and videocall capabilities
- Development of ZK-proof protocols
- Development of SMPC (Secure Multi-Party Computation) protocols
- Development of attribute-based credential protocols.

Technologies used:

- Swift
- Objective C
- CryptoKit

VPN

Designing and developing a Linux kernel module for cryptographic purposes.

Responsibilities:

- Design of the architecture for the solution
- Implement code for elliptic curve cryptography using fiat cryptography (curve25519, p256, p384)
- Implement code for symmetric ciphers using Intel AES instruction set
- Implement code for hash functions (SHA2, Blake2s)
- Implement code for hash-based DRBG according to NIST 800-90A

Technologies used:

- C

Quantum cryptography

This research project aimed at designing and developing cryptographic algorithms on quantum computers.

Responsibilities:

- Implement quantum algorithms for cryptanalysis
- Design new algorithms for quantum key distribution
- Design new algorithms for quantum encryption

Technologies used:

- IBM Q
- qiskit

SCA

This research project studied the security of symmetric ciphers from the perspective of side-channel attacks. The project involved the usage of machine learning algorithms to determine bits from the encryption key using the running time of the cryptographic process.

Responsibilities:

- Research side-channel attacks strategies
- Design and development of new side-channel attacks

Technologies used:

- C
- Tensorflow
- SciKit Learn